

PARKTOWN BOYS' HIGH SCHOOL



June Examination Paper 1

31 May 2024

Subject	Mathematics
Grade	12
Marks	150
Time	3 hours

Examiner

Dr G Nixon

Moderator

Mrs S Rowell

INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This question paper consists of 10 questions, 9 pages and a formula sheet. You may detach the formula sheet for your use.
Please ensure that your question paper is complete.
2. Answer ALL the questions.
3. Number the answers correctly according to the numbering system used in this question paper.
4. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
5. Answers only will NOT necessarily be awarded full marks.
6. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
7. If necessary, round off answers to TWO decimal places, unless stated otherwise.
8. Diagrams are NOT necessarily drawn to scale.
9. Use the mark allocation to guide the length of your solution.
10. Write neatly and legibly.
11. Rule off after each question.

QUESTION 1

1.1 Solve for x :

1.1.1 $(x+3)(x-2)=6$ (3)

1.1.2 $2x^2 + x - 5 = 0$ (correct to TWO decimal places). (3)

1.1.3 $\sqrt{2-x}-4=x$ (4)

1.1.4 $3^x - 3^{1-x} = 2$ (5)

1.1.5 $x^2 + 6x \geq 7$ (4)

1.2 Solve simultaneously for x and y if:

$x - 2y = 1$ and $2x^2 - xy - 5y - 3y^2 - 2 = 0$ (5)

1.3 If 2 and -5 are roots of $ax^2 + bx = 30$, determine the value of $a + b$. (3)

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QUESTION 2

Consider the quadratic sequence: 72; 100; 120; 132; ...

2.1 Determine the n^{th} term, T_n of the sequence. (4)

2.2 Find the maximum value of the sequence. (3)

2.3 Determine which term in the quadratic sequence equals the 12th term of the sequence of first differences. (5)

[12]

QUESTION 3

- 3.1 A geometric sequence has a second term of 6 and its fifth term is 162. Determine the sum to 100 terms of an arithmetic series with the same first term and with a common difference equal to the constant ratio of the given geometric sequence. (5)

- 3.2 Consider the series below:

$$\sum_{k=1}^{\infty} 5 \left(\frac{2-x}{3} \right)^{k-1}$$

- 3.2.1 For which value(s) of x does the series converge? (3)

- 3.2.2 If $x = 0$, determine the least value of m for which:

$$\sum_{k=1}^m 5 \left(\frac{2-x}{3} \right)^{k-1} > 14,99 \quad (5)$$

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QUESTION 4

- 4.1 Given $f(x) = \frac{2}{x+1} - 3$:

- 4.1.1 Determine the coordinates of the y – intercept of f . (2)

- 4.1.2 Find the coordinates of the x – intercept of f . (2)

- 4.1.3 Sketch the graph of f , clearly showing both the asymptotes and the intercepts with the axes. (3)

- 4.1.4 Determine the equation of the decreasing line of symmetry of f . (2)

- 4.2 An increasing exponential function g with equation $g(x) = ab^x + q$ has the following properties:

Range: $y > -3$

The points $(0; -2)$ and $(1; -1)$ lie on the graph of g .

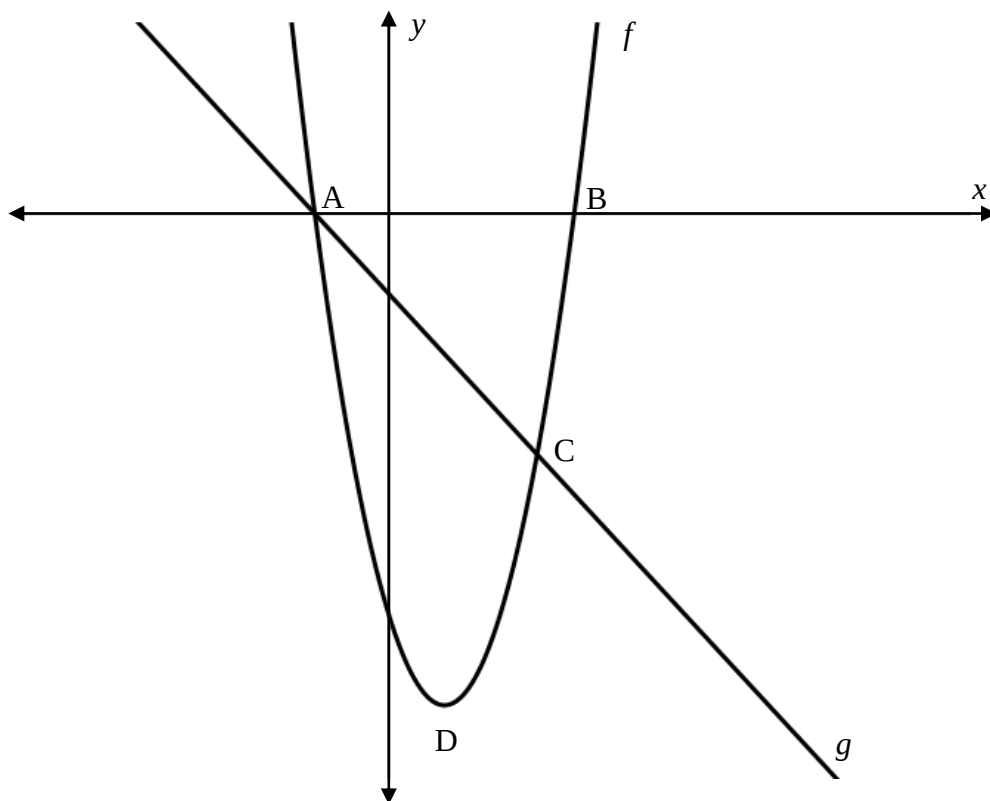
- 4.2.1 Find the equation of g . (4)

- 4.2.2 Describe the translation(s) of g to h where $h(x) = 2 \cdot 2^x + 1$. (2)

[15]

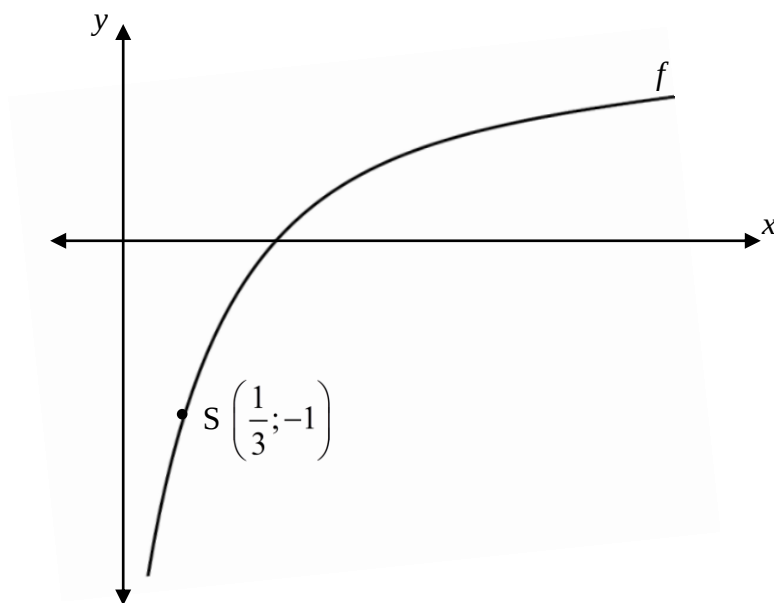
QUESTION 5

- 5.1 $f(x) = 2x^2 - 6x - 20$ and $g(x) = -2x - 4$ are sketched below. A and B are the x -intercepts of f . f and g intersect at A and C. D is the turning point of f .



- 5.1.1 Calculate the coordinates of the turning point D of f . (3)
- 5.1.2 Determine the length of AB where A and B are the x -intercepts of f . (3)
- 5.1.3 Calculate the average gradient of f between A and the point where $x = -1$. (3)
- 5.1.4 Determine the value of p if $2x^2 - 6x + p = 0$ has non-real roots. (2)
- 5.1.5 Determine for which values of x it is true that $f(x) \cdot g(x) > 0$. (2)
- 5.1.6 $h(x)$ is a tangent to f at P. Determine the coordinates of P if h is parallel to g . (4)

5.2 Given $f(x) = \log_a x$ where $a > 0$ and $S\left(\frac{1}{3}; -1\right)$ is a point on the graph of f .



5.2.1 Prove that $a = 3$. (2)

5.2.2 Determine the equation of h , the inverse of f , in the form $y = \dots$ (2)

5.2.3 Find the equation of g if g is the reflection of f in the x – axis. (1)

5.2.4 Find the values of x for which $g(x) \geq 1$. (2)

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QUESTION 6

- 6.1 The annual rate of depreciation of a particular type of machine is 15%, based on the reducing balance method. When the price is R40 000 per machine, after how many years will one of these machines be worth R10 000? (correct to 2 decimal digits). (3)

- 6.2 Peter saves R300 monthly in an account earning interest at 8,5% p.a, compounded monthly. At the end of 10 years he stops making payments but leaves the money to continue earning interest.

Calculate how much Peter can expect to have in the account 20 years after he started his savings plan. (5)

- 6.3 A bank granted Keith a loan of R150 000 at an interest rate of 15,25% per annum compounded quarterly. Keith will repay the loan in 8 equal quarterly payments. Payments begin 3 months after the loan was granted.

6.3.1 Calculate his quarterly payment. (4)

6.3.2 Calculate the balance outstanding immediately after Keith makes his 4th payment. (4)

[16]

QUESTION 7

- 7.1 If $g(x) = x^2 - 2x - 3$, determine $g'(x)$ from first principles. (5)

- 7.2 Find $\frac{dy}{dx}$ for each of the following examples:

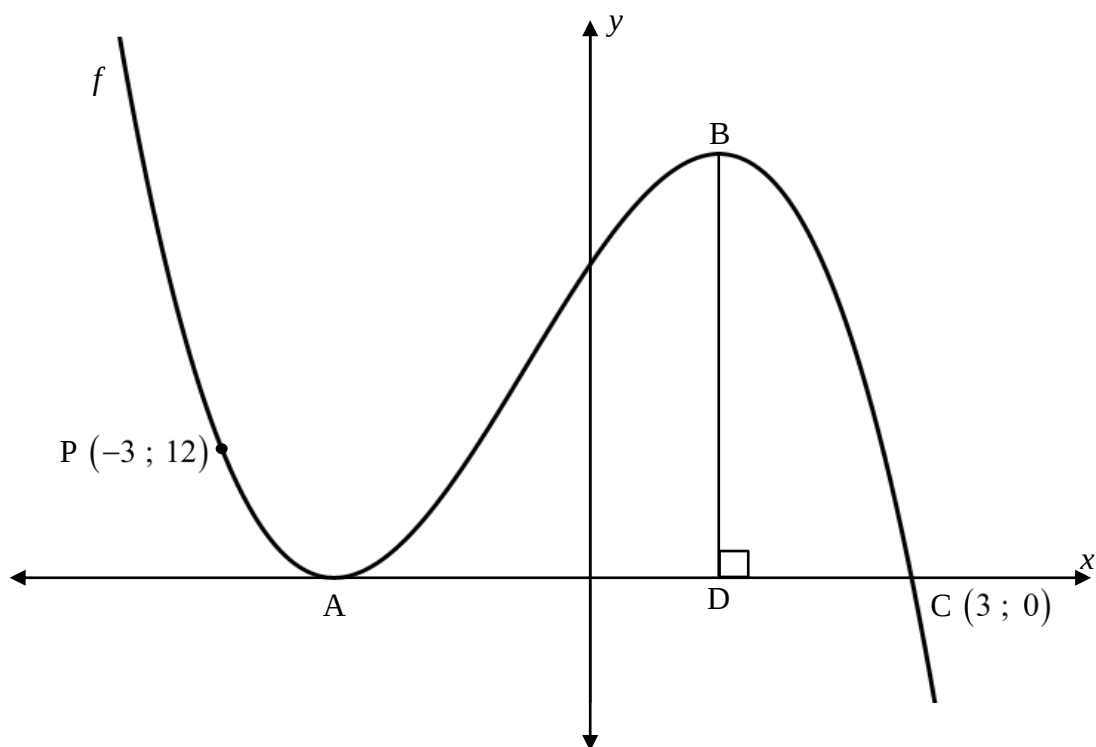
7.2.1 $y = \left(x + \frac{3}{x}\right)\left(x - \frac{3}{x}\right)$ (3)

7.2.2 $y = \frac{x^2 - \sqrt{x^5}}{2x}$ (3)

[11]

QUESTION 8

The sketch below shows the graph of $f(x) = -2x^3 + bx^2 + cx + d$. A and B are the stationary points, A and C(3 ; 0) are the x – intercepts. P(–3 ; 12) is a point on the curve and BD is a line perpendicular to the x – axis at D.



- 8.1 Determine the coordinates of A. (4)
- 8.2 It is further given that $f(x) = -2x^3 - 2x^2 + 16x + 24$.
Determine the length of BD. (5)
- 8.3 Find the equation of the tangent to f at P. (3)
- 8.4 Determine the value(s) of k for which $-2x^3 - 2x^2 + 16x + 24 = k$ has three distinct roots. (2)
- 8.5 Find the x – coordinate of the point of inflection of f . (2)
- 8.6 Determine the value(s) of x for which f is:
- 8.6.1 increasing. (1)
- 8.6.2 concave down. (1)
- 8.7 If $y = f(x) + t$ passes through the origin, determine the value of t . (1)

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QUESTION 9

A and B are exhaustive events where $P(A) = 0,35$ and $P(B) = 0,79$.

- 9.1 Write down the value of $P(A \text{ or } B)$. (1)
- 9.2 Draw a Venn diagram to represent the information. (4)
- 9.3 If $n(S) = 60$, determine the value of $n(A)$. (2)
- [7]**

QUESTION 10

There are seven different shirts and four different pairs of trousers that need to be hung on a rail in a cupboard.

- 10.1 Determine in how many different ways the clothes can be hung on the rail. (1)
- 10.2 Find in how many different ways the clothes can be arranged if all the shirts are to be hung next to each other and all the pairs of trousers need to be hung next to each other on the rail. (2)
- 10.3 Determine the probability that a shirt will hang at the beginning of the rail and a pair of trousers will hang at the end of the rail. (3)
- [6]**